

Synthetic Cultural Agents from Aggregate Anchors

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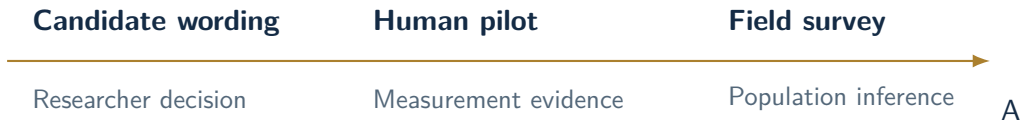
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What does an anchored synthetic policy tell us about a new survey?

Working-paper discussion / 16 September 2026

Before fielding an item

We want to learn whether proposed wording tracks a preference across populations.



synthetic score could inform the first step. Its value must be tested against a subsequent human pilot.

Revised paper, Introduction and Discussion. Pre-pilot usefulness is prospective.

The usual starting point is a persona prompt

“Answer this questionnaire as a typical adult living in Argentina.”

“Respond naturally and sincerely, as someone would in real life. Do not mention being an AI or assistant.”

“Generally speaking, would you say that most people can be trusted or that you need to be very careful in dealing with people?”

1: Most people can be trusted

2: Need to be very careful

“Select exactly one response code from the options below.”

“Return only the response code, with no words or explanation.”

Appendix: evaluation wrapper; PROMPT_SOURCE_EXCERPTS.md: realized Q57 text. Layout reordered; chat control tokens omitted.

Make the supplied population information inspectable

Which information did we supply, and what can the output establish?

Declared construction input

Aggregate preference signs determine labels on a shared comparison bank.

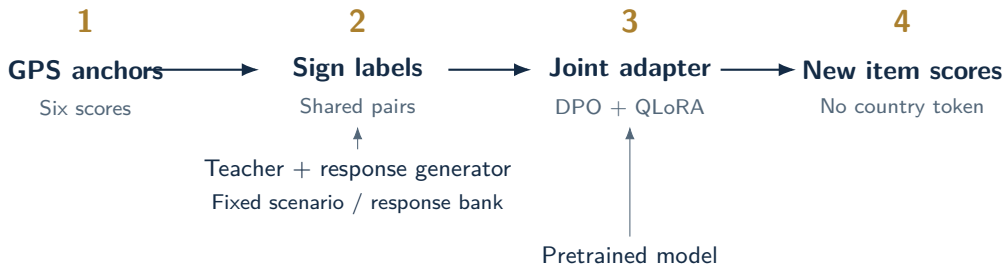
Information that remains

Pretraining, instruction tuning, bank semantics, and fitting variation still affect scores.

Country-unconditioned evaluation removes the direct country-name channel at inference.

Revised paper §§1–3. Cultural fine-tuning and DPO have precedents: CultureLLM; Cao et al.; SubPOP; Jung and Kim. No broad priority claim.

The whole construction, in one view



The new survey enters at readout.

Custom diagram from revised paper §3 and Appendix 7. Production manifest: `use_anchors=false` for optional WVS generation anchors.

Six scores become a sign profile

$$z_c \longrightarrow s_{ck} = \mathbf{1}\{z_{ck} \geq 0\}$$

Argentina: $(-, +, -, +, +, -)$

trust / risk / patience / altruism / pos. recip. / neg. recip.

The United States is non-negative on all six coordinates, so every pair is labeled chosen A. Mexico is negative on all six, so every pair is labeled chosen B.

16 countries, 13 distinct sign profiles. Twins share a labeling problem: Mexico–Russia, India–Greece, Indonesia–Egypt.

Revised paper §3.1 and the sign-profile table. Magnitudes and profile prose do not enter production labels.

Recipe versus invertibility

The map from declared signs through the bank to a policy is a construction.

Identification would invert a named parameter from observed WVS option probabilities.

Adapter-GPS rank association on new text is criterion evidence for that construction.

Whether policies differ only as a function of the assigned sign vector remains a separate question.

Pause Which of those claims should this draft make?

Revised paper §3.1. The design does not recover z , magnitudes, a single GPS coordinate, or culture.

A real synthetic pair receives a deterministic label

Scenario, excerpt: “A fellow student tutored me for free before a critical exam, and now they’re facing a tight deadline for their own project and need someone to review their work.”

A / chosen for Argentina

“I’ll set aside time to carefully review their work this evening.”

B / rejected for Argentina

“I’ll let them know I can only do a quick scan tomorrow morning.”

$$z_{c, \text{positive reciprocity}} = +0.1596793085 \quad \Rightarrow \quad A \succ B$$

Appendix: exact excerpts; complete pair in the Appendix. Intended trait loadings are generator judgments. The label is not an observed Argentine choice.

DPO fits the comparisons; QLoRA limits what moves

A regularized Bradley–Terry pairwise choice criterion

$$\Pr(y_w \succ y_\ell \mid x) = \sigma\left(\beta \left[\log \frac{\pi_\theta(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \log \frac{\pi_\theta(y_\ell \mid x)}{\pi_{\text{ref}}(y_\ell \mid x)} \right]\right)$$

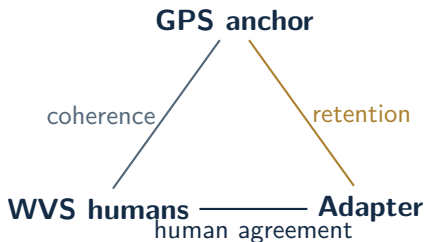
Increase the chosen response's relative likelihood against the rejected response, using the reference policy as the baseline.

QLoRA: keep the quantized Llama 3.1 8B base fixed; fit low-rank adapter weights jointly across dimensions.

526 train / 132 held out rank 16 / beta 0.1 / one epoch

Revised paper §3.2 and Appendix 8; Rafailov et al.; Hu et al.; Dettmers et al. Notebook settings; checkpoint correspondence remains a release requirement.

Evaluation asks three different external questions



Same comparison surface

16 selected countries; 23 common single-choice WVS items; same recodes and aggregation across arms.

Score supplied response codes by normalized log probabilities. No sampled answer is required.

Held-out synthetic comparisons separately ask whether fitting recovers the imposed labels.

Revised paper §§3.3–4. Strict common-support analysis is retrospective. Code likelihood is a scoring convention whose spread need not match population frequencies.

Fitting recovers the imposed synthetic contrasts

0.942

Relative-reward recovery

Chosen–rejected margin improves over the reference.

0.819

Raw choice accuracy

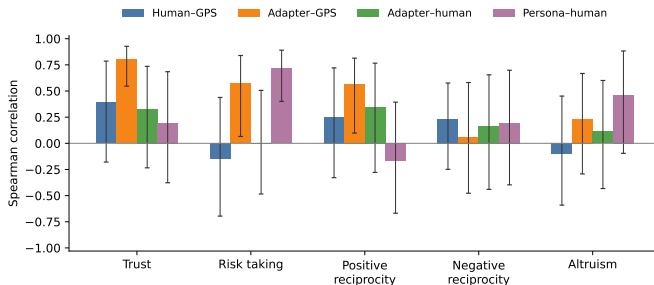
Chosen response has higher adapted-model likelihood.

2,112 held-out country–pair cells reuse a shared bank. Russia has a different holdout split.

Pause What independent test would make this a meaningful first-stage check?

Revised paper §4.1. Recovery is measured against synthetic labels; pooled cells are not independent human choices.

Anchor retention and human agreement differ



Four series, different questions. Risk adapter-human has zero bar height.

Pause Which association should carry the paper's main claim?

Figure 2, unchanged. Country-bootstrap intervals; patience reported itemwise.

Trust retains the anchor; human agreement is uncertain

Country-rank association	ρ	95% interval
Adapter-GPS	0.80	[0.55, 0.93]
Persona-GPS	-0.07	[-0.55, 0.45]
Adapter-human	0.33	[-0.23, 0.74]

Paired adapter-minus-persona contrast: 0.87 [0.22, 1.36].

Permuting GPS trust z across frozen adapter scores, without refitting, gives one-sided $p = 0.002$.

Pause Is anchor-order transfer enough for the contribution we want to make?

Revised paper §4.2 and Appendix C. Intervals omit respondent, anchor, and fitting uncertainty.

A zero does not identify the source of disagreement

Risk / adapter–human

$$\rho = 0$$

95% interval $[-0.48, 0.51]$

Zero sample rank association establishes neither independence nor different latent constructs.

Patience / human–GPS

Q43: work importance -0.56

Q50: financial satisfaction $+0.62$

We retain the coding and interpret these candidate indicators separately.

Pause Which independent evidence would distinguish a mapping problem from a policy limitation?

Revised paper §4.2 and Table 4. Q43 interval $[-0.86, -0.04]$; Q50 $[0.11, 0.91]$.

Configuration sensitivity leaves the mechanism open

Configuration	Trust-GPS ρ	Mean TVD ↓
Base / unconditioned	—	0.462
Base / country persona	-0.07	0.369
Adapter / unconditioned	0.80	0.480
Adapter / country persona	0.67	0.401

The configuration retaining more anchor order need not resemble human response distributions more closely.

Next comparison: give prompting the same anchor information.

Revised paper Table 5: 23 items, 368 country-item cells. TVD is total variation distance. Base bank variation prevents interpreting a base country-ordering coefficient.

Before a journal version

What we would run next, in that order. No new human labels.

- 1 Second scenario bank, same GPS signs, refit two adapters.
- 2 Sign-retrain placebo: permute signs, relabel, and refit.
- 3 Twin-profile check: are Mexico/Russia interchangeable on held-out WVS?

The assignment permutation already in the draft does not refit. Expanding 16 to 24 countries on this bank is later.

Pause If we can only run one of these before submission, which one?

Paper 2 extensions. A blinded pre-pilot remains a later use-test.

What would you need to see before using this?

We can inspect the conditioning rule and observe trust-anchor transfer.

Human agreement and practical pre-pilot value remain unresolved.

Discussion

Which part of the construction is least credible to you?

What evidence would change your view of the proposed use?

Which extension should we run first before a journal version?

Working-paper claim ceiling: group-level scoring instruments intended to complement human survey research.

Appendix. A complete training pair

Scenario. A fellow student tutored me for free before a critical exam, and now they are facing a tight deadline for their own project and need someone to review their work. Decision: I can either take the time to thoroughly review their project or concentrate on my own study schedule. Trade-off: Helping my fellow student would acknowledge the effort they put into tutoring me, but it would cut into my own study time and potentially impact my performance.

A. I will set aside time to carefully review their work this evening. They went out of their way to help me when it mattered most, and I want to make sure they know their effort made a difference. If I do not take this seriously, it would feel like I am ignoring the support that helped me succeed.

B. I will let them know I can only do a quick scan tomorrow morning. My exam prep schedule is already packed with material I have not mastered yet, and I need to be realistic about what I can manage without burning out. I hope they understand my limitations in this situation.

Appendix 7, pair 00023ac74dac. Target: positive reciprocity. Argentina chooses A. Contractions expanded for typesetting.

Appendix. Candidate WVS indicators

Anchor	Items	Candidate interpretation
Trust	Q57–Q64, Q73	Generalized, interpersonal, institutional trust
Patience	Q43; Q50	Less work importance; financial satisfaction
Risk taking	Q106, Q107, Q109, Q178	Economic values; fare avoidance
Positive reciprocity	Q81	Confidence in charitable organizations
Negative reciprocity	Q176, Q177, Q179, Q195	Moral clarity; justifiability
Altruism	Q99, Q101, Q103	Membership in three types of organizations

Composites average item means without range standardization.

Table 2 and Appendix 9. Strict panel excludes Q69–Q71, Q174 and Q12–Q14.

Appendix. How we score response codes

$$\ell_{c q j} = \sum_t \log \hat{\pi}_c(a_{q j t} \mid q, a_{q j, < t}), \quad \hat{P}_c(j \mid q) = \frac{e^{\ell_{c q j}}}{\sum_h e^{\ell_{c q h}}}$$

Candidate strings include a leading space. Sum token log probabilities without length normalization; normalize across the supplied codes.

$$\text{Item score} = \sum_j g_q(j) \hat{P}_c(j \mid q)$$

Human responses use the same recode before survey-weighted averaging. Item scores are then averaged within each proposed composite.

Revised paper §3.3 and Appendix 7. Decoder-only model; teacher-forced scoring requires no sampled response.

Appendix. Uncertainty and remaining gates

Reported uncertainty

Country bootstrap, 2,000 draws, conditional on selected countries, items, and fitted runs. Intervals omit respondent sampling, GPS measurement error, and training randomness.

Bank and Egypt

The shared bank is mixed-trait. Pair loadings are generator judgments. Egypt stays in the primary panel; restoring Q69–Q71 on the other fifteen countries leaves adapter–GPS trust between 0.79 and 0.81. Assignment permutation of GPS trust z , without refitting, gives $p = 0.002$.

Still pending

Pinned checkpoint provenance; isolated calculation replay; independent validation of bank semantics.